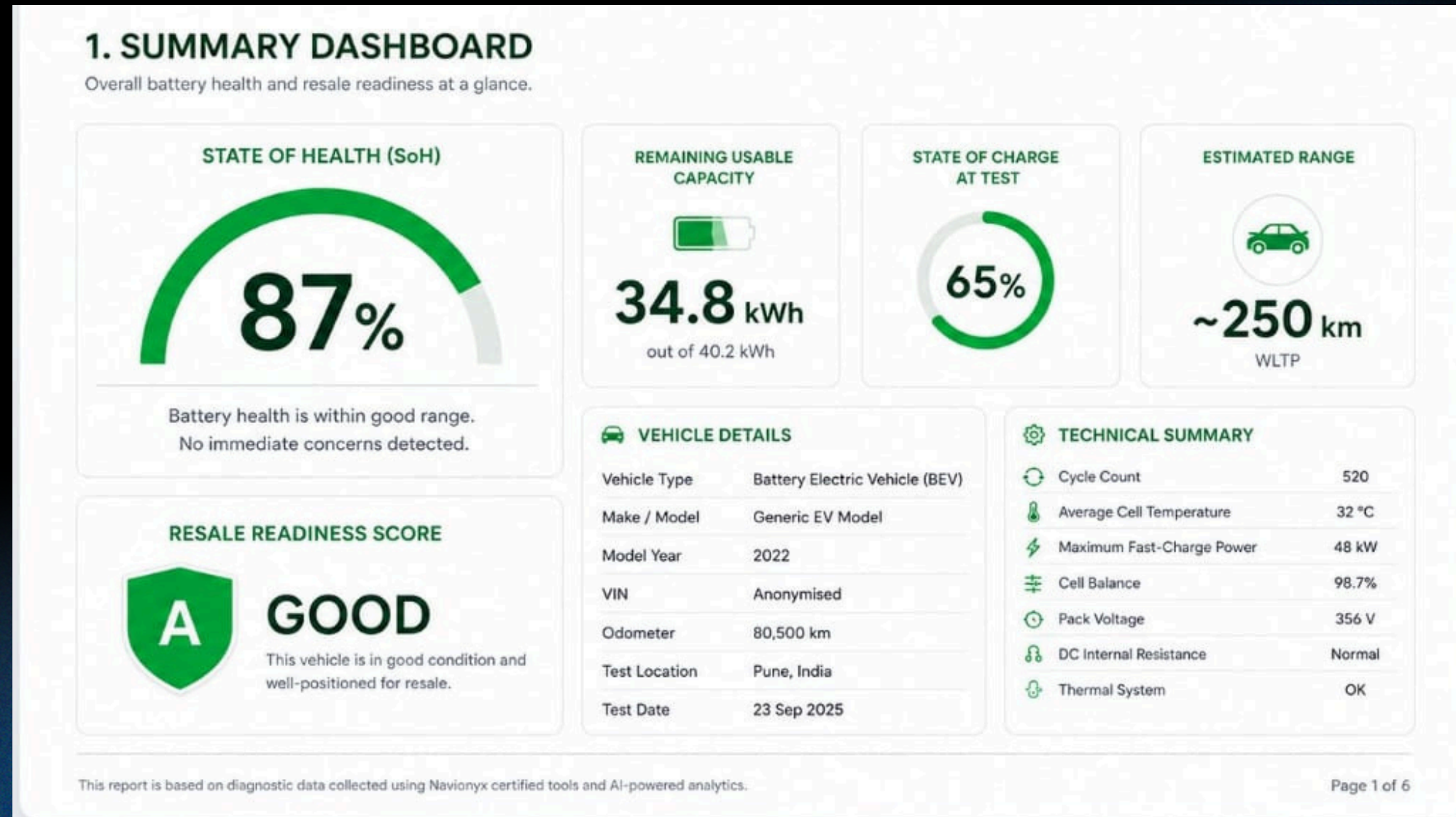


CLEANING PROPOSAL



NAME:- SREEJA SANYAL
WEEK :-2
DAY:-9
DATE:- 09.09.2026

Data Quality Findings:-

The initial data-quality assessment was performed on the battery charging measurement files.

Major observations:-

- Some CSV files contain metadata/header information before the actual measurement records.
- Missing values were observed in different columns/files and need to be investigated before modelling.
- Some files have inconsistent structures or column formats.
- Numerical measurement columns need to be converted into a consistent numerical format.
- Duplicate or repeated records need to be checked.
- Possible abnormal values must be investigated using the physical operating range of the battery.
- The data should be standardized into a common structure before machine-learning modelling.

First 35 lines of the original CSV

```
-----  
1 :  
2 :  
3 : Measurement ID,589  
4 : Battery Name,LG HG2 18650_SN62A4  
5 : Battery Type,Li-ion  
6 : Circuit,Circuit001  
7 : Program,1747 - LG_HG2_CyclesA  
8 : Start Time,11/29/2018 6:52:45 PM  
9 : End Time,12/1/2018 9:57:38 AM  
10 : Test section,Charge1  
11 : Comment,  
12 : OrderNo,  
13 : Producer, LG  
14 : Type, Li-ion  
15 : Nominal Voltage, 3.7  
16 : Nominal Current, 3  
17 : Nominal Capacity, 3  
18 : Cells, 1  
19 : Maximum Voltage, 4.2  
20 : Gassing Voltage, 0  
21 : Break Voltage, 0  
22 : Charge Factor, 0  
23 : Impedance, 0.02  
24 : Cold Cranking Current, 0  
25 : Energy Density, 0  
26 : Comment,HG2 cell with 10AWG wire soldered  
27 :  
28 :
```

screenshot of the original csv file of 0degC
of "589_Charge1.csv"

Why cleaning is required:-

- Data cleaning is necessary to ensure that only valid measurement records are used for analysis and model development.
- Incorrect, duplicated, incomplete or non-measurement rows can introduce bias and may negatively affect model performance.

The main variables of interest are :

- Voltage
- Current
- Temperature
- SOC
- Charging-time-related measurements
- Battery SOC estimation commonly uses measurable quantities such as voltage and current together with suitable algorithms.

```
-----  
DUPLICATE ROW CHECK  
-----
```

File	Duplicate Rows
571_Mixed8.csv	81422
571_Charge13.csv	206
576_UDDS.csv	148898
567_Mixed1.csv	72329
567_Charge6.csv	207
582_LA92.csv	90740
571_Mixed4.csv	73929
571_Cap_1C.csv	368
576_Charge1.csv	130
571_Charge10.csv	208
571_PausCycl.csv	27831
571_Mixed6.csv	73921
575_C20DisCh.csv	2337
571_Mixed5.csv	67174

The duplicated rows present in the dataset which can effectively hinder the performance of model.

Operating Regions, Target Coverage

1. Operating regions to be identified

The cleaned dataset will be analysed to determine the normal operating regions of:

- Battery Voltage
- Charging Current
- Temperature
- State of Charge (SOC)
- Charging/elapsed time

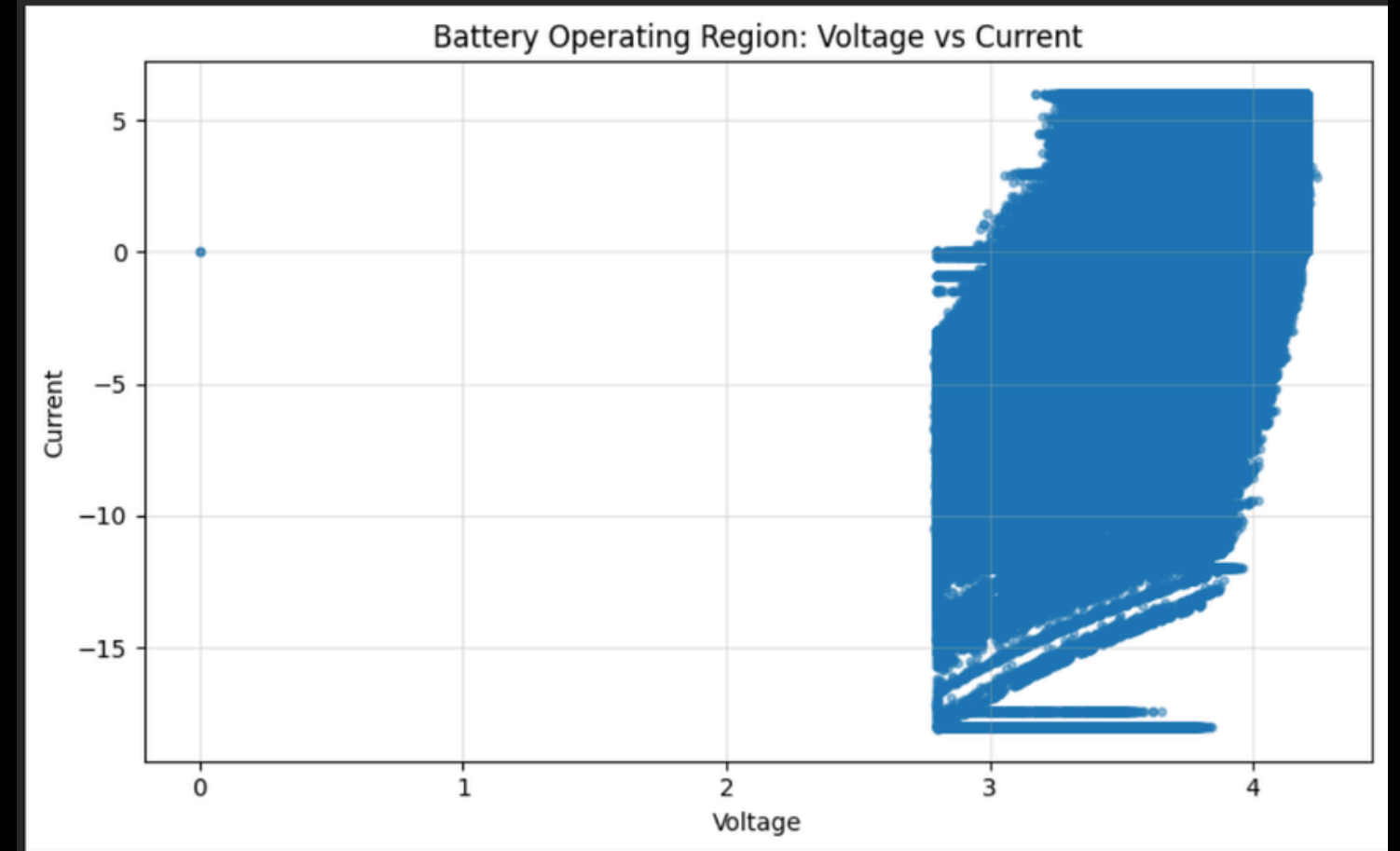
The distributions and relationships between these variables will be visualized to identify normal operating regions and potentially abnormal measurements.

2. Target regions

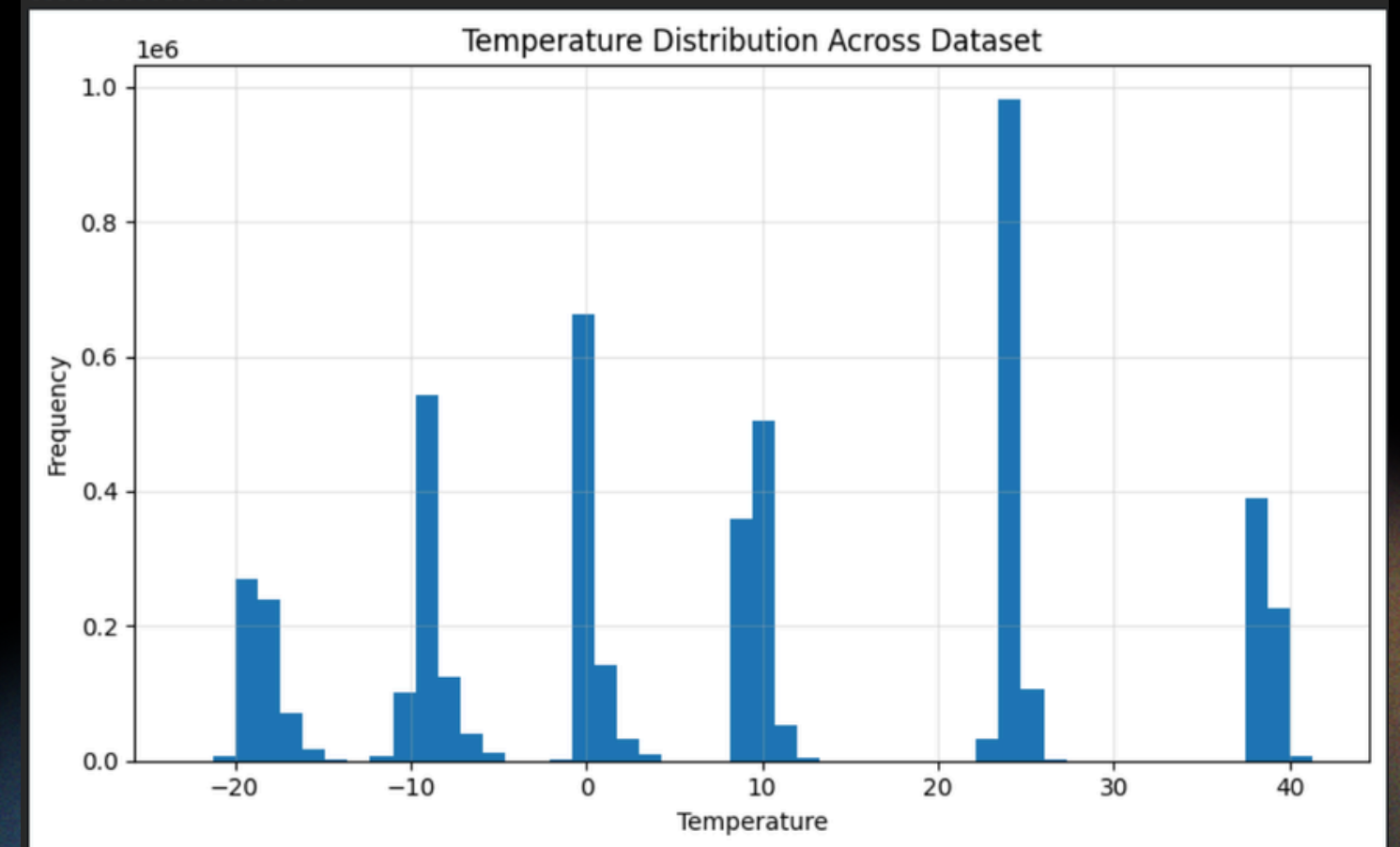
The two main prediction targets are:

- Battery State of Charge (SOC)
- Remaining Charging Time

21. GRAPH 1 - VOLTAGE VS CURRENT



24. GRAPH 4 - TEMPERATURE DISTRIBUTION



Leakage risk

Data leakage is a major concern because battery measurements are sequential and may belong to the same charging cycle/session.

Potential leakage can occur if:

- Information from the future is used to predict an earlier measurement.
- A target-derived variable is accidentally used as an input feature.
- Measurements from the same charging cycle appear in both training and testing data.
- Random row-wise splitting allows highly similar consecutive measurements to occur in both sets.

POSSIBLE DATA LEAKAGE CHECK				
File	Feature	Possible Leakage		Reason
571_Mixed8.csv	Unnamed: 0	False	No obvious leakage from column name	
571_Charge13.csv	Unnamed: 0	False	No obvious leakage from column name	
576_UDDS.csv	Unnamed: 0	False	No obvious leakage from column name	
567_Mixed1.csv	Unnamed: 0	False	No obvious leakage from column name	
567_Charge6.csv	Unnamed: 0	False	No obvious leakage from column name	
582_LA92.csv	Unnamed: 0	False	No obvious leakage from column name	
571_Mixed4.csv	Unnamed: 0	False	No obvious leakage from column name	
571_Cap_1C.csv	Unnamed: 0	False	No obvious leakage from column name	
576_Charge1.csv	Unnamed: 0	False	No obvious leakage from column name	
571_Charge10.csv	Unnamed: 0	False	No obvious leakage from column name	
571_PausCycl.csv	Unnamed: 0	False	No obvious leakage from column name	
571_Mixed6.csv	Unnamed: 0	False	No obvious leakage from column name	
575_C20DisCh.csv	Unnamed: 0	False	No obvious leakage from column name	
571_Mixed5.csv	Unnamed: 0	False	No obvious leakage from column name	
567_Mixed2.csv	Unnamed: 0	False	No obvious leakage from column name	
575_Dis_0p5C.csv	Unnamed: 0	False	No obvious leakage from column name	
571_Charge16.csv	Unnamed: 0	False	No obvious leakage from column name	
576_Cap_1C.csv	Unnamed: 0	False	No obvious leakage from column name	
571_Charge11.csv	Unnamed: 0	False	No obvious leakage from column name	
576_HWFET.csv	Unnamed: 0	False	No obvious leakage from column name	
575_Dis_2C.csv	Unnamed: 0	False	No obvious leakage from column name	
576_Charge3.csv	Unnamed: 0	False	No obvious leakage from column name	

Proposed approach

Use cycle/session-aware data splitting wherever the dataset structure allows it. Features should only contain information that would realistically be available at the prediction time.

This is important because independent cycle-level testing can help prevent leakage and provide a more realistic evaluation of generalization.

Proposed Cleaning Plan

<u>Cleaning requirement</u>	<u>Proposed action</u>
Metadata and inconsistent columns	Remove metadata rows and standardize column names
Non-numeric values	Convert valid values and flag invalid entries
Missing values	Quantify and investigate before treatment
Duplicates	Detect and remove confirmed duplicates
Abnormal measurements	Flag values outside expected operating ranges
Target leakage	Remove future- or target-derived features
Train/test contamination	Use cycle- or session-aware splitting
Different file structures	Convert files to a common schema

THANK YOU!